

COMMENTS: EM-BASED LIKELIHOOD INFERENCE FOR SOME LIFETIME DISTRIBUTIONS BASED ON LEFT TRUNCATED AND RIGHT CENSORED DATA AND ASSOCIATED MODEL DISCRIMINATION

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This paper gives a nice discussion covering many interesting and important aspects of left truncated and right censored (LTRC) data modelling and analyses. One aspect not covered by the paper is the estimation of the underlying distributions involved and especially that of the lifetime. These have important applications in inter alia, construction of goodness-of-fit statistics, an area not well researched for LTRC data. In the following some of these aspects will be discussed briefly. As an example, the estimated distribution function and Q-Q plots will be given for simulated data sets considered by the authors in previous papers.

The notation of the paper will be used. Let Y , C , τ^L denote respectively the lifetime, the censoring time and the time to truncation and let $X = \min(Y, C)$. Let δ and v denote the censoring and truncation indicators, respectively. Furthermore, let F_Y denote the lifetime distribution function. This distribution function is of particular interest.

A nonparametric maximum likelihood estimator F_n for F_Y was proposed in Tsai, Jewel and Wang (1986) (see also Wang, Jewel and Tsai, 1986) as

$$F_n(y) = 1 - \prod_{i: X_i \leq y} \left[1 - \frac{1}{nC_n(X_i)} \right]^{\delta_i}.$$

Here $nC_n(x) = \sum I(\tau_i^L \leq x \leq X_i)$.

Note that F_n reduces to the Kaplan-Meier product-limit estimator when there is no truncation and to the Lynden-Bell (1971) estimator when there is no censoring.

Building on the proposal of Tsai et al. (1986), a number of authors have subsequently derived various properties of this estimator and proposed variations of it. Some of their results will now be discussed.

Gijbels and Wang (1993) obtained a strong i.i.d. representation for $F_Y(y)$ of order $O(n^{-1} \log n)$. This result was then applied to obtain strong consistency and asymptotic normality of a kernel density estimator of the density of F_Y . See also a slight extension of these results by Zhou (1996).

By considering the product-limit process $Z_n(y) = \sqrt{n}[F_n(y) - F_Y(y)]$, Tse (2003) obtained a uniform strong approximation of Z_n of order $O(n^{-1/6} \log n)$, over fixed intervals, by a Gaussian process. This immediately leads to weak convergence results for $Z_n(y)$.

In statistical inference one is often also interested in the quantile function. A natural estimator for $Q_Y(u) \equiv F_Y^{-1}(u) = \inf\{y : F_Y(y) \geq u\}$ is the product-limit quantile function given by

$$Q_n(u) \equiv F_n^{-1}(u) = \inf\{y : F_n(y) \geq u\}.$$

In the case where Q_Y is a continuous function, it is perhaps more natural to use a smoothed estimator rather than a step function such as Q_n . For this Zhou, Sun and Ren (2000) defined a kernel quantile function estimator based on Q_n . They proved its asymptotic normality and also derived a Berry-Esseen type bound for this approximation. For the product-limit quantile function Q_n Tse (2005) obtained results for the corresponding quantile process similar to those that he obtained in Tse (2003). This then also leads to weak convergence results for the quantile process.

An important application of estimators of the lifetime distribution and quantile functions is in the construction of goodness-of-fit statistics. Well-known (omnibus) test statistics such as those of Kolmogorov-Smirnov, Cramer-von Mises and Anderson-Darling are all based on the empirical and/or the quantile functions (see e.g., D'Agostino and Stephens, 1986). The above mentioned results on the weak convergence of the empirical and quantile processes, can be used to find the limiting distributions for the mentioned test statistics. The limiting distribution of course contains unknown quantities that have to be estimated in order to find the critical values. A simpler, although computationally more challenging, approach to finding such critical values is to make use of the bootstrap. Gross and Lai (1996) has given an asymptotic justification for the bootstrap for LTRC data and have shown that it gives an approximation to the sampling distribution of order $O(n^{-1})$ for various nonparametric statistics. The application to goodness-of-fit statistics of the product-limit estimators of the distribution and quantile functions need to be further investigated, using both asymptotic results and the bootstrap to obtain the needed critical values.

An important aspect that has not yet been addressed in these papers is what the effect of estimated parameters in the underlying model is on the limiting distributions. In the usual empirical and quantile processes this effect has been studied intensively. Similar results in the LTRC model seem to be lacking. Such results will be needed if the limiting distributions are to be used in testing goodness-of-fit.

An extension of the LTRC model to the case of covariates seems particularly relevant. In a recent paper Shen (2014) studied this using the quantile regression approach of Koenker and Bassett (1978) (see also Koenker, 2005). He defined the quantile regression estimator for the conditional quantile of the lifetime given a fixed value of the covariate vector and derived its asymptotic normality. These results are particularly useful when one wants to study the lifetime quantiles at different levels rather than at a single “mean” value. More complex quantile regression models need to be studied further.

Clearly there are a number of areas for further research in this very exciting and relevant area of Statistics. The current paper by Balakrishnan and Mitra will certainly stimulate such interest amongst readers of this journal.

Example

This illustrative example shows an estimate of the distribution function and also the form of the resulting Q-Q plots in case of a correct distribution assumption and an incorrect assumption.

In a previous paper (Balakrishnan and Mitra, 2012) the authors give a simulated data set of size 100 from a Weibull distribution, with a 40% truncation. Estimating the lifetime distribution function using the above estimator, gives the result in Figure 1.

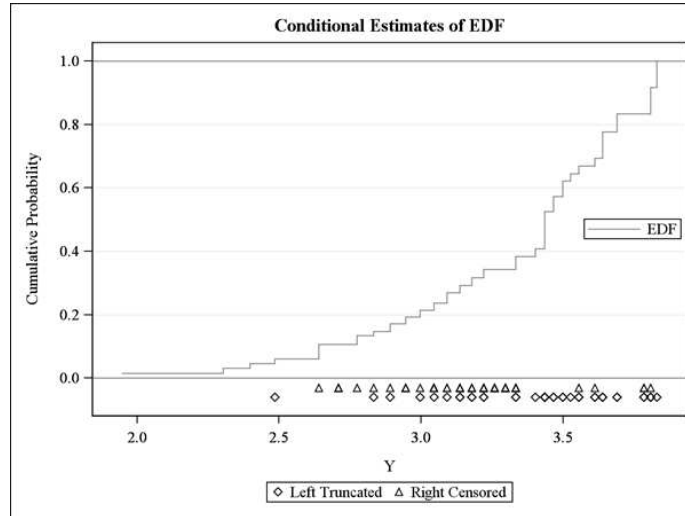


Figure 1: Estimated distribution function for Weibull data.

Inverting the estimated distribution function gives an empirical quantile function from which a Q-Q plot can be drawn. This is given in Figure 2 for the Weibull data using the estimated parameter values given by the authors. As expected, it is reasonably straight lined.

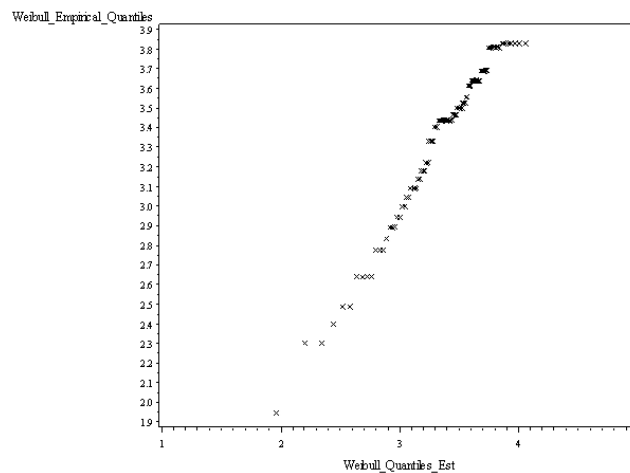


Figure 2: Q-Q plot of Weibull vs Weibull quantiles.

If it is assumed that the underlying distribution is Weibull while in fact it is, for example, gamma, a distinctly non-straight line Q-Q plot should result. In Balakrishnan and Mitra (2013) a simulated data set of size 100 from a gamma distribution is given. From this the empirical quantile function was estimated. A Q-Q plot was then drawn, assuming the underlying distribution is Weibull, i.e. an incorrect assumption. The result is given in Figure 3, which is clearly curved as expected.

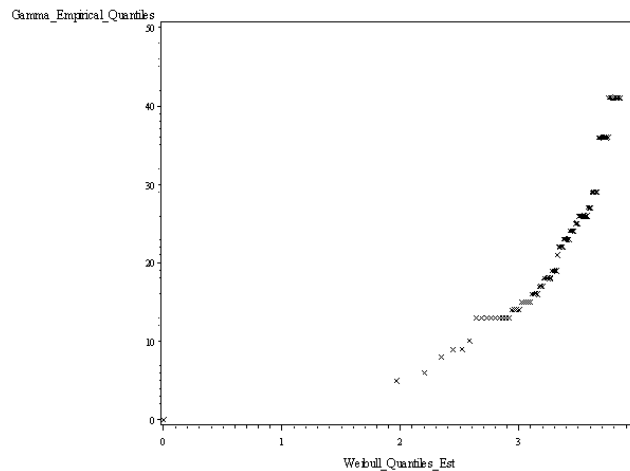


Figure 3: Q-Q plot of Gamma vs Weibull quantiles.

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